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INGRESS PREDICTION AND DETECTION WITH INTELLIGENT VEHICLE RESPONSE AND WHEEL APPARATUS

Abstract

Two or more sensors positioned at a wheel of a stopped vehicle may detect an unwanted substance, and transmit signals, based on which a height level of the substance is determined. The system may also determine a type of the substance, such as water, snow, sand or mud, based on electrical resistance of the substance detected by the sensors. Cameras or other imaging equipment, including LIDAR, may also be used to determine the substance height and/or composition. Accordingly, actions to protect the vehicle may be taken, such as raising or lowering the ride height of the vehicle body using an adjustable suspension of the vehicle. Other protective actions may include closing vents, seals, valves and battery isolation units, and transmitting messages to a vehicle driver and/or to emergency personnel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 18/529,609, filed on Dec. 5, 2023. The disclosures of the referenced application are hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] The present disclosure relates to taking actions to protect a vehicle, such as a parked vehicle, from water, snow, sand or other accumulations of substances at the vehicle, based on detection of the accumulations of substances at the wheels.

SUMMARY

[0003] Parked vehicles face a variety of risks, particularly when left in an unattended state. Flooding, deep snow, sand storms, and mud can damage a vehicle or can accumulate under or around the vehicle and get the vehicle stuck.

[0004] In one approach, under certain conditions, using the suspension system of the vehicle the ride height of the vehicle is raised so that the vehicle body is elevated farther away from the ground. This may be used, for example, when the vehicle is wading through water or muddy terrain. However, the vehicle may be more vulnerable when the vehicle is stopped as water or other substances can gradually accumulate around. Also, a parked vehicle may be unattended and unmonitored. In addition, a vehicle body may be raised in response to detection of water during wading but the vehicle body would never be lowered below normal ride height in response to wading. Further, while the vehicle is wading through water, a sensor on a wheel would be impractical because the wheel is constantly spinning and thus the location of the detection of the water would be unknown. Moreover, during wading, water is being swept up by the wheels and being splashed by other parts of the vehicle. Thus, a sensor on a wheel may not yield useful information regarding the location of the detection of the water at the wheel.

[0005] According to another approach, water level sensors are positioned on a body of the vehicle, for example, to detect an ambient water level. But such water level sensors are typically positioned too high on the vehicle to be able to provide early warning of flooding or other material building up around the vehicle. Often sensors are positioned so high above ground level that a flood or rising water can cause damage to the vehicle. Also, sensors often provide a binary signal: either a relatively high water level is detected to trigger a wading operation by the vehicle, or no water level signal is provided. Sensors may be positioned in a wheel well but a detection by such sensors may not provide early warning about an accumulating unwanted substance.

[0006] Also, a sensor may detect water but may not discern the type of substance accumulating around the vehicle. Without this type of granular information, it may be impossible to determine what kind of protective action is appropriate for the vehicle. For example, a high water level at the vehicle may warrant elevating the ride height of the vehicle body to avoid flood damage, while a high snow level at the vehicle may warrant lowering the vehicle to avoid getting stuck in snow accumulation beneath the vehicle body.

[0007] According to an aspect of the disclosure as described herein, sensors may be positioned at one or more wheels of a vehicle to detect water or other material buildup at the vehicle. The sensors may detect the height level of the water or other buildup, and may determine the type of substance—water, snow, ice, mud, sand, dust, or the like. The sensors may detect electrical conductivity and/or temperature of the buildup and, accordingly, the sensors may transmit signals, for example, nearfield wireless radio frequency signals, such as Bluetooth, to a data processor

onboard the vehicle, such as to an electronic control unit (ECU) of the vehicle.

[0008] In response to the substance height determination and the substance composition determination, the system may instruct one or more protective actions for the vehicle. Such protective actions may include, for example, using the suspension system of the vehicle to elevate or to lower the ride height of the body of the vehicle, or portions thereof, relative to a current ride height of the vehicle body or relative to a normal parked ride height of the vehicle body. The system may in addition, or instead, command further protective actions, including one or more of closing/opening one or more vehicle vents, such as radiator air vents, shutting off/turning on one or more vehicle systems, for example, heating or cooling systems, closing/opening one or more valves, such as heater valves, disconnecting one or more batteries, transmitting a message requesting intervention by a vehicle driver/owner and/or by emergency personnel, and the like.

[0009] Two or more sensors may be positioned on, or in, a wheel rim (e.g., equidistant from a center of the wheel), or may be positioned on or at the hubcap of the wheel. Each sensor may include a pair of probes to measure electrical conductivity of the substance at the wheel. Sensors may also include a thermometer or other temperature sensor, a camera, charge couple device (CCD), spectrometer, LIDAR or other imaging equipment, to help identify the height level of the substance at the wheel and the type of substance at the wheel. According to the height level determined, the system may then transmit command to vehicle components to take one or more of the previously discussed or other protective actions.

[0010] Facilitating ongoing determination of height level of substances, the sensors may continue to monitor conditions and may continue to transmit data about the conditions. A first sensor may detect a substance at a first time and a second sensor may detect no substance at the first time. At a subsequent time, both the first and the second sensors may detect the substance. Based on such data, the system may determine that the wheel of the stopped vehicle is positioned such that the first sensor is lower than the second sensor. Since the distance between sensors is known in advance, the rate of accumulation of the substance may be determined. The sensors may also determine whether the buildup is at more than one wheel of the vehicle, and thus determine whether the substance is accumulating only locally, if only the sensors at one wheel detect accumulation, or at all the wheels. Knowing which wheel's sensors have detected the unwanted substance may allow the system to decide appropriate protective actions. For example, snow may accumulate on a first side of a vehicle parked at a curb as snow is shoveled off the sidewalk on the first side of the vehicle. In response to sensors on the first side detecting unwanted substance accumulation and the ECU determining that the snow is likely snow, the ECU may command the suspension systems at the wheels on the first side of the vehicle to lower the vehicle body to prevent further snow accumulation under the vehicle. At the same time, if sensors at the wheels on the second side of the vehicle determine no substantial snow level, then the ECU may maintain the suspension systems at the wheels on the second side of the vehicle at a normal height. By not lowering the second side of the vehicle body, the vehicle body may get less dirty as other vehicles pass on the second side of the vehicle and splash dirty wet snow and salt that had been deposited on the road surface (to prevent the snow freezing to the pavement).

[0011] Based on the signals received from the sensors, the onboard data processor may determine the height level of the buildup of the substance in contact with the wheel(s) and may determine the composition of the buildup material. The composition of the substance may be determined based on sensor data including electrical conductivity of the substance, a temperature of the substance, a rate of accumulation of the substance, photographic image analysis of the substance, the variation in the substance as detected at various sensors of the vehicle and at various times, weather and/or other news report data and/or other such data. Identifying the substance type may allow the ECU to make better decisions regarding what protective action(s) to take. For example, if the substance is determined to be water then the ECU may command raising the vehicle body to prevent or to reduce damage to the battery, the engine, other components and systems, and the vehicle body

itself. On the other hand, if the ECU determines that the substance is snow or sand, then the ECU may command lowering the vehicle body to prevent accumulation under the vehicle that could interfere with wheel traction or that could contact the bottom of the vehicle body to ground the vehicle.

[0012] Disclosed are an apparatus, system, method, means for implementing the method, and a non-transitory computer-readable medium comprising instructions that, when executed by a processor, implement the method. The method may include receiving signals from two or more sensors positioned on a wheel of a stopped vehicle; based on the received signals, a height level of a substance in contact with the wheel may be determined; also based on the received signals, the system may determine a type of the substance; and then the system may determine a protective ride height for at least a portion of the vehicle based on the height level of the substance and the type of the substance. An adjustable suspension of the vehicle may then adjust a current ride height of the vehicle to the protective ride height.

[0013] In one configuration, a first sensor of the two or more sensors is positioned at a first radial distance from a center of the wheel, and a second sensor of the two or more sensors may be positioned at the same first radial distance from the center of the wheel and spaced remote from the first sensor. In this configuration, the determining of the height level of the substance in contact with the wheel may entail determining that the substance is in contact with both the first and second sensors. The determining the protective ride height may include determining, based on the type of substance in contact with the wheel being water, a raised ride height relative to a normal stopped ride height. A third sensor may also be positioned at the same first radial distance from the center of the wheel and spaced remote from the first sensor and from the second sensor. The determining of the height level of the substance in contact with the wheel may then entail determining that the substance is in contact with the third sensor, with the second sensor and with the first sensor, and in this case the ride height may be raised to a highest position obtained by the adjustable suspension. In addition, if it is first determined that the substance is in contact with the first sensor and that the second sensor is free of contact with the substance, and later it is determined that the first sensor and the second sensor are in contact with the substance, then an accumulation rate of the substance may be calculated based on a time interval between the first time and the second time. Based on the accumulation rate, a message may be transmitted to a device, such a mobile telephone of a driver or emergency personnel, remote from the vehicle.

[0014] In the configuration in which the first and second sensors are each positioned at a first radial distance from a center of the wheel and spaced remote from each other, the height level of the substance in contact with the wheel may be determined by determining contact of the substance with the first sensor and second sensors, and the determining the protective ride height may entail determining, based on the type of substance in contact with the wheel being snow or sand, a lowered ride height relative to a normal stopped ride height. The lowered ride height may be a lowest position obtained by the adjustable suspension. The two or more sensors may be positioned at a rim of the wheel.

[0015] Also contemplated are an apparatus, system, method, means for implementing the method, and a non-transitory computer-readable medium comprising instructions that, when executed by a processor, implement the method. The apparatus may include a vehicle wheel that has a first sensor positioned at a rim of the wheel at a first distance from a center of the wheel. The first sensor may include first probes that can detect electrical conductivity of a substance in contact with the wheel, and a second sensor also positioned at the rim of the wheel at the first distance from the center of the wheel and spaced remote from the first sensor and also having probes that can detect the electrical conductivity of the substance. A wireless transmitter may be connected to the first and second sensors and may transmit signals indicating the electrical conductivity detected by the first sensor and the electrical conductivity detected by the second sensor.

[0016] For example, the first sensor and the second sensor each may include a temperature sensor

and/or other sensors. An orientation sensor may also be included at the wheel to detect an orientation of the wheel in its current rotational position.

[0017] A third sensor may be additionally positioned at the rim of the wheel at the same first distance from the center of the wheel and spaced equidistant from the first sensor and from the second sensor. The first sensor may also be positioned equidistant from the second sensor and from the third sensor. The third sensor may also include probes that can detect the electrical conductivity of the substance in contact with the wheel. In an embodiment, one probe of one or more of the sensors may be positioned at an inner ring of the rim of the wheel, and another probe of that sensor may be positioned at an outer ring of the rim of the wheel.

[0018] In an embodiment, an onboard data processor, such as an electronic control unit of the vehicle, may be positioned and configured to receive the signals indicating the electrical conductivity detected by the first sensor and the electrical conductivity detected by the second sensor. Then, the onboard data processor may determine a protective action for the vehicle based on the signals received.

[0019] Other aspects and features of the present disclosure will become apparent to those ordinarily skilled in the art upon review of the following description of specific embodiments in conjunction with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The present disclosure, in accordance with one or more various embodiments, is described in detail with reference to the following figures. The drawings are provided for purposes of illustration only and merely depict typical or example embodiments. These drawings are provided to facilitate an understanding of the concepts disclosed herein and should not be considered limiting of the breadth, scope, or applicability of these concepts. It should be noted that for clarity and ease of illustration, these drawings are not necessarily made to scale.

[0021] FIG. 1 illustrates an example of a process for detecting an unwanted substance (UWS) at a vehicle and for taking steps to protect the vehicle, including the options to raise or to lower the vehicle body, and possible additional protective actions, according to an aspect of the disclosure;

[0022] FIGS. 2A-2C illustrate, respectively, a raised ride height, normal parked ride height and lowered ride height of a vehicle, according to an aspect of the disclosure;

[0023] FIG. 3 illustrates an electronic control unit (ECU) of a vehicle together with other vehicle systems, and also shows nodes in communication with the ECU, according to an aspect of the disclosure;

[0024] FIGS. 4A-4E illustrate a vehicle wheel and unwanted substance detectors thereon, according to an aspect of the disclosure;

[0025] FIG. 5 illustrates a system according to an aspect of the disclosure;

[0026] FIGS. 6A and 6B illustrate the ECU, suspension systems, further sensors, and other components of the vehicle, according to an aspect of the disclosure;

[0027] FIG. 7 is a flowchart showing an operation of the system, according to an aspect of the disclosure.

DETAILED DESCRIPTION

[0028] It will be appreciated that for simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood that the embodiments and examples described herein may be practiced without these specific details. In other instances, well-known methods, procedures and components, including software, firmware

and hardware components, have not been described in detail so as not to obscure the embodiments described herein. Also, the description is not to be considered as limiting the scope of the embodiments described herein.

[0029] FIG. **1** illustrates an example of a process for determining an unwanted substance and taking steps for protecting a vehicle **201**, according to an aspect of the present disclosure. While depicted in some of the figures as a sedan, the term vehicle may encompass other types of automobiles, SUVs, vans, buses, trucks, tractors, motorcycles, including hybrid and electric vehicles, or the like. Some of the discussion describes a vehicle with four wheels, however, vehicles with more or fewer than four wheels are also contemplated. The term parked may sometimes be used to refer to the state of the vehicle, however, it will be understood that the vehicle may be stopped or slowed or moving without departing from the spirit of the disclosure.

[0030] A process as illustrated in FIG. **1** may be commenced automatically in response to one or more sensor signals indicating that a sensor at one or more wheels has detected an unwanted substance. A parked car, for an example, an electric vehicle, with the engine or motor turned on or off, may keep one or more sensors always turned on. Each sensor may be powered by its own battery or all sensors of or on a wheel may share the same power source. A smaller number of sensors, for example one sensor, or all sensors of a wheel, for example, at only one wheel of a four wheel vehicle, may be left on at all times, and detection by the sensor(s) of an unwanted substance (UWS) may trigger it to transmit a notification to an onboard processor, such as an electronic control unit (ECU), which, in turn, may transmit notification to the other sensors at the wheels to “wake up” and begin monitoring for unwanted substances (UWS), or the notification from the initial sensor(s) may be transmitted directly to the remaining sensor so as to wake up the remaining sensors.

[0031] Aspects of the disclosure will now be described with reference to FIGS. **1** and **2A-2C**. As shown at **102** of FIG. **1**, one or more sensors positioned at a wheel **401** of vehicle **201** illustrated in FIGS. **2A-2C**, may monitor for detection of an unwanted substance (UWS). Detection of an UWS by a first UWS sensor **431** may cause it to alert other UWS sensors by direct transmission to sensors or indirectly, by way of transmission to onboard ECU **301**, which may then transmit a wake up signal to other UWS sensors. From this point on, other UWS sensors would also be monitoring for the UWS.

[0032] At **104**, the ECU **401** may receive data from one or more UWS sensors. The UWS sensors may transmit the data periodically or upon detection of a threshold level of UWS. Or the UWS sensors may transmit the data in response to request of the ECU **401**. The ECU **401** evaluates the sensor data and may make a separate determination of the height level of the UWS for each UWS sensor. The ECU **401** may determine the composition of the UWS after or substantially contemporaneously with determining the height level of the UWS, or may make the composition determination only if the height level of the UWS is determined to be a threshold height. For example, a threshold height may be a height in which more than one sensor at a wheel detects the presence of UWS. Or a threshold may be a specific height level from the ground, for example 10-20 centimeters from the ground.

[0033] At **106**, the ECU **401** determines whether UWS is detected. In an embodiment, the ECU **401** determines that the UWS is present and takes action only if the threshold height level is determined. If no substantial UWS is detected then processing returns to **102**. Otherwise processing continues to **108**.

[0034] At **108**, the ECU **401** may determine whether the vehicle **201** is parked. For example, the vehicle **201** may be stopped but not parked, as indicated by the gear shift stick being in drive gear or in a gear other than parked, as indicated by activation of a brake pedal, or as indicated by recent vehicle motion accessed by the ECU in an electronic vehicle history. Or, the vehicle may be in motion and one or more sensors may have detected UWS. In an embodiment, the sensors are disabled when the vehicle **201** is not in parked state. In a further embodiment, the sensors and the

vehicle protection system are enabled automatically when the ignition and/or primary electric motor is off and/or when the vehicle is determined to be free of a driver and passengers. The ECU **301** may determine that the vehicle is free of the driver and passengers based on seat weight sensors, seat belt buckle latched/open sensors, cameras or the like, vehicle doors opening/closing or the like.

[0035] In an embodiment, if the vehicle is not parked, then no further automatic action may be required because the driver is likely present and may take protective actions for the vehicle. Thus, if at **108** the vehicle is determined not to be parked, then processing continues to **110**; otherwise (“no” at **108**), processing returns to **102** if the vehicle is determined to be parked.

[0036] At **110**, the ECU **401** determines a type of vehicle protection action warranted based on the determination by the ECU **401** of the height of the UWS and based on the determination by the ECU **401** of the composition of the UWS. The ECU **301** may determine the ride height adjustment, by controlling the suspension systems of the vehicle, which may be warranted for the vehicle, as shown in FIGS. **1** and **2A-2C**. In embodiment, if the type of substance cannot be determined based on the data from the sensor, then by default the action may be to raise the vehicle body. For example, one or more cameras may be out of order or out of focus. In an embodiment, if the UWS height level and the type of substance is not determined then the vehicle body is always raised to a maximum height.

[0037] At **112**, the system may transmit a signal to the adjustable suspension system at one or more of the wheels to raise the vehicle body. The adjustable suspension systems at each wheel of the vehicle may work independently to raise or to lower the vehicle body, and thus the vehicle body may be thought of as having four parts that may be raised or lowered. That is, if the suspension system at the left front wheel alone raises the vehicle body, then the left front portion of the vehicle body is raised while the rest of the vehicle body remains where it was. The system may command raising the vehicle body in response to a determination by the ECU that the height of the unwanted substance detected at the wheel or wheels may damage one or more components or systems of the vehicle or the vehicle body, for example, in response to a determination that the substance is water.

[0038] At **114**, the adjustable suspension systems at each wheel, or at fewer than all the wheels, may raise the vehicle body to an elevated ride height. Monitoring may then return to **102**.

According to an embodiment, if after a period of time, for example 1-90 minutes, no unwanted substance is detected by the sensors at the vehicle wheels, or if no unwanted substance at a threshold height level that would trigger the raising of the suspension or the taking of other protective action is determined by the sensors at the vehicle wheels, then the system may command that the suspension system return the ride height to the normal parking ride height.

[0039] On the other hand, if at **110** the system determines that the ride height of the vehicle body, or one or more portions thereof, is to be lowered, then processing proceeds to **116**. At **116**, the system instructs lowering the vehicle body or one or more portions thereof.

[0040] At **118**, the adjustable suspension systems lower the vehicle body. In an embodiment, one portion of the vehicle body, for example the first side of the vehicle body, may be raised, while a second side of the vehicle body may be lowered. For example, this may be desirable if people are shoveling snow from the sidewalk to the second side of the vehicle, while traffic continues on the first side of the vehicle which might splash dirt or road salt onto the first side.

[0041] FIG. **3** is a schematic illustration of components of the vehicle **201**, including the electronic control unit ECU **301**. One or more of components of the vehicle **201**, including controllers **311-321**, may be integrated with the ECU **301** or may be logically connected thereto. Communication between the ECU **301** and one or more other components, including the unwanted substance sensors (UWS) **331**, may be via wireless or wired connection. FIG. **3** shows transceiver **333** connected to the UWS sensors **331** for communicating with the ECU **301** and/or with one or more devices, for example, with a portable device, such as a mobile telephone of the driver. Each UWS sensor **331** may be integrated with, or be connected to, a transceiver **333**, or all the UWS sensors

331 of a wheel may share the same transceiver **333**. Thus, for example, in a vehicle with four wheels, there may be four such transceivers **333**. Or, only two of the wheels of the vehicle, that is, one wheel on each side of the vehicle, may have UWS sensors, and all sensors on each wheel may share a transceiver **333**.

[0042] As shown in FIG. 3, the vehicle **201** may include an internal and external seal controller **311**, a battery isolation controller **313**, an ignition controller **315**, an adaptive suspension controller **317**, a vent controller **319**, and an ingress prevention mode controller **321**, which control respective actuators and systems of the vehicle **201**. The ingress prevention mode controller **321** may be a module of the ECU **301** that manages vent controller **319**, battery isolation controller **313**, and/or internal and external seal controller **311**.

[0043] UWS sensors **431-449** may be positioned at one or more wheels **401a-401d** of the vehicle **201** and may transmit to the ECU **301** data regarding UWS at each sensor. Each UWS sensor **431, 433, 435, 437, 439, 441, 443, 445, 447, 449** may include, or may be connected to, a radio frequency transceiver, such as a near-field transceiver, for example, a Bluetooth transceiver, for communicating with the ECU **301**. Or, several UWS sensors may be connected and share an RF transceiver for communicating with the ECU **301**. Periodically, upon request for data by the ECU **301**, upon a UWS **331** detecting UWS, or upon the occurrence of other events, one or more UWS sensors may transmit UWS detection data to the ECU **301**. While 10 UWS sensors are illustrated for the wheel **401**, more than 10 or fewer than 10 sensors may be provided. Some sensors may be positioned in the inner rim of the wheel **401** and some on the outer rim of the wheel **401**. In an embodiment, the sensors may be positioned substantially equidistant from the center of the wheel **401** and equidistant from each other to facilitate the ECU **301** determining height level of the UWS.

[0044] The ECU **301** may also receive external data information, for example, from emergency band radio tuner **335**, citizens' band (CB), and/or from network communications module **337**, which may be connected to a network such as the internet. The emergency band radio tuner **335**, and/or the network communications module **337** that communicates with a server **371** via a network **363**, may receive weather or other environmental conditions information, including coastal, river or storm flooding, watermain breaks, avalanches, tsunamis or the like, weather forecast or weather report information, such as an impending snowstorm, sandstorm, or like.

[0045] In an embodiment, the ECU **301** may activate the sensors **431-449** automatically in response to notification, for example, of coastal flooding expected for an area in which the vehicle **201** is located. A machine learning model may be used to predict that a weather forecast or report indicates a snow, flooding or other such condition for an area in which the vehicle is located. The ECU **301** may determine that the vehicle **201** is located in the area based on GPS data. In a further embodiment, if the ECU **301** is notified, for example, that a major snowstorm is expected, then the ECU **301** may take protective action earlier in response to detection of snow by the UWS sensors. For example, the ECU **301** may lower the vehicle body upon detection of snow by any one UWS sensor, without waiting for confirmation of the snow height level and/or of the composition of the UWS by a second or third UWS sensor. Or, upon such notification, the ECU **301** may determine by default that the composition of the UWS detected by the sensors is snow, unless the ECU **301** makes a positive determination that the UWS comprises some other substance.

[0046] The detectors may be turned on in response to a user turning them on in anticipation of potential threats, for example, based on weather forecasts or other news reports. In a further embodiment, the onboard system may activate the sensors **431-449**, or a remote node, such as server **371** or mobile device **373**, may make the determination that a weather/news report that indicates a threat condition and may then transmit a notification of the threat condition via network **363** and network communications module **337** to the ECU **301** of the vehicle **201**. For example, a machine learning model may be used to predict that a weather forecast or report indicates heavy snow, flooding or some other such threat condition for an area in which the vehicle is located. The

server **371** or mobile device **373** may know the location of the vehicle **201** because the ECU **201** may transmit its location to the server **371** or mobile device **373**, or the server **371** or mobile device **373** may transmit the notification of the threat condition based on the general area, for example, the city or postal code, in which the vehicle **201** is registered or with which the vehicle is associated. Based on such a notification received by the ECU **301**, the ECU **301** may wake up the sensors and/or may take the other precautionary actions described above.

[0047] Seal actuators **341**, also shown in FIG. **3**, may be controlled by internal and external seal controller **311**. By way of further example, one or more seals at or around the battery or the exhaust system may be controlled by seal actuator **341**.

[0048] Also shown in FIG. **3** are vent actuators **343** that may be controlled by vent controller **319**. One or more events, such as a radiator vent, air conditioner vent or duct, and/or exhaust vent, may be opened or closed by vent actuators **343**. Windows of the vehicle determined to be open may also be closed in this way.

[0049] As discussed, adaptive suspension systems **345a-d** may be used to raise or to lower the ride height of the vehicle body. Adaptive suspension systems **345a-d** may be controlled by adaptive suspension controller **317**.

[0050] Battery isolation unit **347**, also shown in FIG. **3**, may isolate one or more batteries of the vehicle using various means, for example, by closing or sealing a battery compartment. One or more batteries may also be disconnected for safety at this time. Battery isolation unit **347** may be controlled by battery isolation controller **313**.

[0051] FIGS. **4A-C** illustrate a vehicle wheel **401** with a rim **403**, a hubcap **405** and spokes **409**. As shown in FIG. **4A**, one or more sensors **431-449** may be positioned at detection points along the rim **403** or at other portions of the wheel **401**.

[0052] Sensors **431-449** may each comprise a pair of electrical probes and be positioned on the rim **403** equidistant from each other at an equal radius from the center of the wheel **401**, as shown in FIG. **4A**. Each sensor may include electrical probes **431a**, **431b**, shown by way of example in FIG. **4C** as being located at the outer ring of the rim **403** and the inner ring of the rim **403**, respectively. The electrical probes **431a**, **431b** may measure a change in electrical resistance between them. The change in electrical resistance may indicate the type of substance or the composition of the substance that is in contact with the wheel. For example, water is electrically conductive, and the increased conductivity compared with air may be detected by the probes **431a-431b**. Snow is a poor electrical conductor. In this way, the composition of the substance may be determined based on the signal detected by the sensors. More or fewer sensors than the number illustrated may be provided. More than one wheel may have sensors and each wheel need not have the same number of sensors. Also, while shown as being deployed at the rim **403** of the wheel **401**, in addition or instead, one or more sensors may be provided elsewhere in or on the wheel **401**, for example, on the spokes **409** and/or on the hubcap **405**. Other types of sensors, including cameras or charge coupled devices, thermometers and temperature sensors, pressure sensors, weight sensors, and the like, as well as combinations of more than one type of sensors, may be provided on or at the wheel, or on or at more than one wheel, to detect substance height levels and/or to distinguish between various types of substance compositions.

[0053] The ECU **301** may determine the height level of the UWS by determining an orientation of the wheel **401** based on which UWS sensor first signals contact with the UWS. The UWS sensor that first signals contact with the UWS may be presumed to be the lowest UWS sensor on the wheel **401** in the wheel's current rotational position. In the illustration shown in FIG. **4A** this may be UWS sensor **433**. As the UWS accumulates and rises, one or more additional UWS sensors may signal contact with the UWS, as shown in FIG. **4A**.

[0054] In an embodiment, to determine a height level of the UWS, an orientation sensor may also be provided to determine an orientation of the wheel **401**. Based on data received from the orientation sensor, the system may determine the height of the UWS sensors in contact with the

wheel given the current position of the wheel. The system could then determine the height level of the substance at the wheel, since the location on the wheel of the UWS sensor that detects the UWS is known in advance, and the orientation sensor of the wheel may aid the ECU **301** to locate the UWS sensor in relation to the ground. A level sensor may also be integrated with or connected to each UWS sensor to aid the ECU **301** determine the position of the UWS sensor on the wheel given the current rotational position of the wheel.

[0055] Further, based on the electrical resistance of the substance determined by the probes the ECU **301** may determine the composition of the UWS. In an embodiment, photography or other light sensor data may be collected by cameras to aid the system in determining the composition of the UWS and/or weight or pressure of the substance as measured by the sensors and/or the temperature of the substance measured at the sensors, may be used by the system to determine that the substance is water or primarily water. Laser imaging, detection, and ranging (LIDAR) or other imaging detecting may be used to determine the height level of the UWS and the composition of the UWS.

[0056] As shown in FIG. **4C**, the UWS sensor **431** may include an electrical conductivity sensor, which may include two prongs **431a**, **431b** separated by a gap. The electrical conductivity of the UWS in the gap may be determined by the resistance detected between the prongs **431a**, **431b**. Pairs of prongs or probes of each sensor may be positioned in, on or at the rim, as shown in FIG. **4A**. In an embodiment, as shown in FIG. **4C** the first prong or probe **431a** may be positioned in, on or at an outer ring of the rim **403** of the wheel **401**, and the second prong or probe of the sensor **431** may be positioned in, on or at the inner ring of the rim **403** of the wheel **401**. One or more sensors may also be positioned at other portions of the wheel **401**, such as the hubcap **405** or one or more of the spokes **409**. The sensor **431** may incorporate a thermometer or other temperature sensor. The sensor **431** may include a camera or charge couple device (CCD) that captures photographs or detects other image data of the UWS. Such data may be useful for determining a high level of the UWS and/or for determining the composition of the UWS.

[0057] FIGS. **4D** and **4E** illustrate two rotational positions of the same wheel **401** with two UWS height levels: at a first time T.sub.1 and at a subsequent second time T.sub.2. UWS sensor **431** is in contact with the UWS at the first time T.sub.1 both when the wheel **401** is in the rotational position shown in FIG. **4D** and when the wheel **401** is in the rotational position shown in FIG. **4E**. However, when the wheel **401** is in the rotational position shown in FIG. **4D**, the UWS sensor **431** contacts the UWS earlier at a time before time T.sub.1. At second time T₂, on the other hand, in the rotational position shown in FIG. **4D** only the UWS sensor **431** is in contact with the UWS while in the rotational position shown in FIG. **4E** both the UWS sensors **431** and **439** are in contact with the UWS. Thus, in the rotational position shown in FIG. **4E**, first contact with the UWS is signaled later but contact by a second sensor is signaled earlier than in the rotational position shown in FIG. **4D**. Due to such uncertainties, the height level of the UWS at the time of first contact with the first UWS sensor has to be determined by calculating D₁, the height of the UWS sensor **431** relative to the ground given the current rotational position of the wheel **401**.

[0058] The rotational position of the wheel **410** may be determined because the ECU **301** may be fed a reading from an orientation sensor of the wheel **401**, which may show a deviation of the rotational position from a first rotational position of the wheel **401**. The ECU **301** also has access to a memory storing the length of the radius of the wheel **401**, which is the distance from the center **411** of the wheel **401** to the ground, assuming the tire is withing a normal tire pressure range. Thus, a vertical line segment R with a length equal to the known radius of the wheel **401** may be drawn from the center of the wheel to the portion **413** of the wheel at the ground, as shown in FIG. **4E**. The line segment S with known length may be drawn between the UWS sensor **431** and the center **411** of the wheel **401**, since the distance between the UWS sensor **431** and the center **411** of the wheel **401** is known in advance. Based on the known rotational position of the wheel **401**, the ECU **301** may determine the angle α between the line segment R and the line segment S. The line

segment S plus a line segment D2, as shown in FIG. 4E, form the hypotenuse of a right triangle that is formed with the vertical line segment R and a line segment D3. The line segment D3 is drawn orthogonal to R connecting to the end of this hypotenuse. The length of the line segment D2 is thus equal to length of $R/\cos(\alpha)$ –length of S. Thus, based on the angle α and the length of line segment R, the length of the hypotenuse may be calculated.

[0059] The line segment D1, which is equal to the height of the UWS sensor **431** relative to the ground, may then be obtained as follows: D2 is the hypotenuse of a smaller triangle formed with D1 and a portion of D3 opposite the angle β . The angles α and β are equal, since D1 is parallel to R, and both D1 and R intersect the line segment S. The length of the line segment D1 may thus be calculated as (length of D2)×cosine(α). In this way, the height of the UWS level may be obtained because the radius of the wheel and the distance S of each sensor from the center **411** are known in advance, and the orientation of the wheel may be determined based on an indication received from the orientation sensor.

[0060] FIG. 5 illustrates an example of an implementation of a system **500**, which may comprise an ECU **301** and some components thereof, that may receive vehicle sensor data and driver input and may transmit instructions to various components and systems of the vehicle **201** to control the motion sequences described herein.

[0061] System **500** may receive content and data via input/output (I/O) path **502** that may comprise I/O circuitry (e.g., network card, or wireless transceiver). I/O path **502** may provide content (e.g., broadcast programming, on-demand programming, Internet content, content available over a local area network (LAN) or wide area network (WAN), and/or other content) and data to control circuitry **504**, which may comprise processing circuitry **506** and storage **508**. Control circuitry **504** may be used to send and receive commands, requests, and other suitable data using I/O path **502**, which may comprise I/O circuitry (sometimes referred to as communication circuitry). I/O path **502** may connect control circuitry **504** (and specifically processing circuitry **506**) to one or more communications paths (described below). I/O functions may be provided by one or more of these communications paths, but are shown as a single path in FIG. 5 to avoid overcomplicating the drawing.

[0062] Control circuitry **504** may be based on any suitable circuitry and may include processing circuitry **506**. As referred to herein, control circuitry should be understood to mean circuitry based on one or more microprocessors, microcontrollers, digital signal processors, programmable logic devices, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), etc., and may include a multi-core processor (e.g., dual-core, quad-core, hexa-core, or any suitable number of cores) or supercomputer. In some embodiments, control circuitry may be distributed across multiple separate processors or processing units, for example, multiple of the same type of processing units (e.g., two Intel Core i9 processors) or multiple different processors (e.g., an Intel Core i7 processor and an Intel Core i9 processor). In some embodiments, control circuitry **504** executes instructions for an AR application stored in memory (e.g., storage **508**).

[0063] In client/server-based embodiments, control circuitry **504** may include communications circuitry suitable for communicating with a server or other networks or servers. Computing device **500** may be a part of a local area network with one or more of devices **500** or may be part of a cloud computing environment accessed via the internet. In a cloud computing environment, various types of computing services for performing searches on the internet or informational databases, providing storage (e.g., for a database) or parsing data (e.g., using machine learning algorithms described above and below) are provided by a collection of network-accessible computing and storage resources (sometimes referred to as “the cloud”).

[0064] Control circuitry **504** may include communications circuitry suitable for communicating with a server, edge computing systems and devices, a table or database server, or other networks or servers. The instructions for carrying out the above mentioned functionality may be stored on a server. Communications may involve the Internet or any other suitable communication networks or

paths. In addition, communications circuitry may include circuitry that enables peer-to-peer communication of user equipment devices, or communication of user equipment devices in locations remote from each other (described in more detail herein).

[0065] Memory may be an electronic storage device provided as storage **508** that is part of control circuitry **504**. As referred to herein, the phrase “electronic storage device” or “storage device” should be understood to mean any device for storing electronic data, computer software, or firmware, such as random-access memory, read-only memory, hard drives, optical drives, digital video disc (DVD) recorders, compact disc (CD) recorders, BLU-RAY disc (BD) recorders, BLU-RAY 3D disc recorders, digital video recorders, solid state devices, quantum storage devices, gaming consoles, or any other suitable fixed or removable storage devices, and/or any combination of the same. Nonvolatile memory may also be used (e.g., to launch a boot-up routine and other instructions). Cloud-based storage, described in relation to FIG. 5, may be used to supplement storage **508** or instead of storage **508**.

[0066] Control circuitry **504** may include video and audio and/or audible output generating circuitry and tuning circuitry. Encoding circuitry (e.g., for converting over-the-air, analog, or digital signals to MPEG signals for storage) may also be provided. Control circuitry **504** may also include scaler circuitry for upconverting and down converting content into the preferred output format of system **500**. Control circuitry **504** may also include digital-to-analog converter circuitry and analog-to-digital converter circuitry for converting between digital and analog signals. The tuning and encoding circuitry may be used by system **500**, **501** to receive and to display, to play, or to record information, such as sensor data. The circuitry described herein, including for example, the tuning, video generating, encoding, decoding, encrypting, decrypting, scaler, and analog/digital circuitry, may be implemented using software running on one or more general purpose or specialized processors.

[0067] Control circuitry **504** may receive instruction from a user by way of user input interface **510**. User input interface **510** may be any suitable user interface, such as dashboard controllers, an onboard command console or touch screen, a touchpad, stylus input, joystick, voice recognition interface, or other user input interfaces. Display **512** may be provided as a stand-alone device or integrated with other elements of each one of system **500** and devices **501**. For example, display **512** may be a touchscreen or touch-sensitive display. In such circumstances, user input interface **510** may be integrated with or combined with display **512**. In some embodiments, user input interface **510** includes a remote-control device having one or more microphones, buttons, keypads, any other components configured to receive user input or combinations thereof. For example, user input interface **510** may include a handheld remote-control device having an alphanumeric keypad and option buttons. In a further example, user input interface **510** may include a handheld remote-control device having a microphone and control circuitry configured to receive and identify voice commands and transmit information.

[0068] Audio output equipment **514** may be integrated with or combined with display **512**. Display **512** may be one or more of a monitor, a television, a liquid crystal display (LCD) for a mobile device, amorphous silicon display, low-temperature polysilicon display, electronic ink display, electrophoretic display, active matrix display, electro-wetting display, electro-fluidic display, cathode ray tube display, light-emitting diode display, electroluminescent display, plasma display panel, high-performance addressing display, thin-film transistor display, organic light-emitting diode display, surface-conduction electron-emitter display (SED), laser television, carbon nanotubes, quantum dot display, interferometric modulator display, or any other suitable equipment for displaying visual images. A video card or graphics card may generate the output to the display **512**. Audio output equipment **514** may be provided as integrated with other elements of each one of device **500** and equipment **501** or may be stand-alone units. An audio component of videos and other content displayed on display **512** may be played through speakers (or headphones) of audio output equipment **514**.

[0069] An application and/or any instructions for performing any of the embodiments discussed herein may be encoded on computer-readable media. Computer-readable media includes any media capable of storing data. The computer-readable media may be non-transitory including, but not limited to, volatile and non-volatile computer memory or storage devices such as a hard disk, floppy disk, USB drive, DVD, CD, media card, register memory, processor cache, Random Access Memory (RAM), etc.

[0070] Data for use by a thick or thin client implemented on each one of system **500** and devices **501** may be retrieved on-demand by issuing requests to a server remote to each one of system **500** and user equipment devices **501**. For example, the remote server may store the instructions for the application in a storage device. The remote server may process the stored instructions using circuitry (e.g., control circuitry **504**) and generate the displays discussed above and below. The client device may receive the displays generated by the remote server and may display the content of the displays locally on device **500**. In this way, the processing of the instructions is performed remotely by the server while the resulting displays (e.g., that may include text, a keyboard, or other visuals) are provided locally on device **500**.

[0071] Control circuitry may be based on any suitable control circuitry such as one or more microprocessors, microcontrollers, digital signal processors, programmable logic devices, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), etc., and may include a multi-core processor (e.g., dual-core, quad-core, hexa-core, or any suitable number of cores) or supercomputer. In some embodiments, control circuitry **511** may be distributed across multiple separate processors or processing units, for example, multiple of the same type of processing units (e.g., two Intel Core i9 processors) or multiple different processors (e.g., an Intel Core i7 processor and an Intel Core i9 processor). In some embodiments, control circuitry executes instructions for an emulation system application stored in memory (e.g., the storage). Memory may be an electronic storage device provided as storage that is part of control circuitry.

[0072] FIGS. **6A-6B** show an example of an ECU **301** and vehicle components and sensors according to an aspect of the present disclosure. The suspension systems may include airbag chambers and respective weight sensor **621**, **623**, **625**, **627** connected to the ECU **301**. The suspension systems **611**, **613**, **615**, **617** may be controlled by corresponding independence suspension and steering systems **651**, **653**, **655**, **657**, respectively, which may be connected to the ECU **301** and managed by independent wheel controller-suspensions (ISCS) **602**.

[0073] A variety of sensors may also be positioned at various parts of the vehicle, including at a side or underside of the vehicle body, and/or at each wheel, including front left wheel weight sensor **621** and front left wheel moisture sensor **641** provided at or near front left wheel **111**, as well as depth sensing cameras **631a**, **631b**. Similarly, front right wheel weight sensor **623** and front right wheel moisture sensor **643**, as well as depth sensing cameras **633a**, **633b** may be provided at or near right front wheel, left rear wheel weight sensor and left rear wheel moisture sensor, as well as depth sensing cameras **635a**, **635b** may be provided at or near left rear wheel **115**, and right rear wheel weight sensor **627** and right rear wheel moisture sensor **647**, as well as depth sensing cameras **637a**, **637b** may be provided at or near right rear wheel.

[0074] Additional sensors, such as additional depth sensing cameras, may be positioned at or near the wheels and/or at various portions of the vehicle, such as at the underside of the vehicle body, for example, air pressure sensors for each wheel (not shown). FIGS. **6A-6B** also illustrate other sensors that may also be connected to, or provided integrally formed, with ECU **301**, including level sensor **607**. FIGS. **6A-6B** illustrate air intake and air release lines for controlling air pressure in each wheel. The air intake and air release lines for each wheel are controlled by air pressure control system **661**, **663**, **665**, **667** for respective wheels of the vehicle, connected to air tank **603** provided with compressor **604** and a pressure sensor **605** for managing each of the air inflate and air release lines. They may be fed readings from the pressure sensor **605** to control the compressor **605** and the air tank **603** as well as the air pressure control system **661**, **663**, **665**, **667** for respective

suspension systems at the wheels. The UWS sensors **331** comprising electrical resistance sensors are not illustrated in FIGS. **6A-6B**.

[0075] FIG. **7** is a flowchart that illustrates a process according to an aspect of the disclosure. The method **700** may be implemented, in whole or in part, by the system **501** shown in FIG. **5** working in concert with other vehicle components illustrated in FIGS. **3**, **6A** and **6B**. One or more actions of the method **700** may be incorporated into or combined with one or more actions of any other process or embodiments described herein. The method **700**, or portions thereof, may be saved to a memory or storage (e.g., of the systems shown in FIG. **5**) or locally as one or more instructions or routines, which may be executed by any suitable device or system having access to the memory or storage to implement the method **700**.

[0076] As shown in FIG. **7**, at **702** the system may monitor for a signal from one or more UWS sensors **331** indicating detection of a substance in contact with sensors **331**. The system may continue to monitor even after the engine or main motor of the vehicle **201** has been turned off and the driver has left and locked the vehicle. In an embodiment, turning off the engine or main motor, and/or locking the vehicle, and/or detecting that the vehicle is unoccupied, automatically initiates the monitoring.

[0077] As shown in FIG. **7**, at **704** the system receives from one or more UWS sensors **331** signaling indicating detection of a substance in contact with sensors **331**. The UWS sensors may communicate with the ECU **301** via transceiver **333**. Data from one or more sensors may be received based on which the ECU may determine whether the sensors signals indicate contact with a substance. If not, then processing returns to **702**.

[0078] If the ECU **301** determines that the sensors signals indicate contact with a substance, then at **706** the ECU **301** may determine a height level of the substance. Since the ECU **301** knows in advance the positions of the sensors **331** and the distance between them, the ECU **301** may determine the approximate height level of the substance, an example of which is illustrated in FIG. **4**. Also, the ECU **301** may store in memory the time at which each UWS sensor transmits its signal and then the ECU **301** may calculate a rate at which the substance is accumulating based on the time at which additional UWS sensors on the wheel **401** subsequently report contact with the substance. Based on the rate of accumulation, the ECU **301** may determine how dire the situation is or is likely to be and take additional protective measures, such as contacting the driver's mobile device **373** via network **363** communicating with the ECU **301** through network communications module **337**.

[0079] At **708**, the ECU **301** may determine a composition of the substance, or at least may determine primary ingredients of the substance. For example if the substance is muddy water, the ECU may determine that the substance is water. The ECU **301** may request data from additional sensors to aid in determining the height level of the substance and/or to aid in determining the composition of the substance. Moisture sensors **641** at the wheels may determine that the wheels are wet, which may facilitate the ECU **301** in determining that the substance is water or another liquid. Cameras **631a-637e** may capture photographic or other image data to help identify the substance. A light source (not shown) may be provided to illuminate the area at which the cameras are directed. The cameras may capture images in the IR, UV, RF, visible, microwave and/or other frequency bands.

[0080] A machine learning-based module may determine a height level of the UWS, and/or identify the substance in the image(s) captured. In an embodiment, the system may first determine a composition of the UWS and then determine a height level, as knowing the composition of the UWS may aid the image analysis process determine the height level of the substance. The machine learning-based model may be trained in advance and may be located off-vehicle with the ECU **301** transmitting LIDAR or other imaging data in real time to the server **371** via network communications module **337** and network **363** and receiving real time predictions from the model regarding UWS height level and/or UWS composition.

[0081] A thermometer or other temperature detecting equipment may aid in identifying the type of substance. For example, if it is determined that the temperature of the substance is significantly less than 0 degrees C. than the system can rule out water in a liquid state (assuming road salt or other chemicals have not been added to lower the melting temperature of the substance).

[0082] At **710**, the system may determine whether, based on the positions of the sensors that report contact with the substance, the substance reaches a height level that warrants protective action. The threshold high level may be different for various substances detected. For example, when a height level of 10 cm of snow is detected this may be the threshold in response to which the ECU **301** takes the action of lowering the ride height of the vehicle body. On the other hand, when a height level of 10 cm of water is detected this may be under the threshold in response to which the ECU **301** takes any action. If the threshold height level for the substance is not detected, then processing returns to **702**.

[0083] If yes, then at **712**, the system may determine an appropriate protective action or set of actions. When a threshold height for snow or sand is detected, then the ECU **301** may determine that suspension systems **621**, **623**, **625**, **627** of the vehicle will lower the vehicle body. Lowering the ride height of the vehicle body may prevent the tires of the vehicle from getting stuck in the substance or grounding the vehicle. When a threshold height for water is determined, then the ECU **301** may determine that suspension systems **621**, **623**, **625**, **627** are to elevate the ride height of the vehicle body.

[0084] At **714**, the ECU **301** transmits instructions to implement such actions. The ECU **301** may communicate with adaptive suspension controller **317** to request that the adaptive suspension systems **345a-d** at each wheel lower the vehicle body or elevate the vehicle body from a normal or current ride height. The adjusting of the ride height may be a preset amount according to the height level of the substance detected. A maximal ride height or a lowest ride height may be commanded depending on the detected substance height level. In addition to, or instead of, instructions for taking steps to adjust the ride height of the vehicle body, the ECU **301** may instruct other protective actions for the vehicle in responsive to the substance detection. ECU **301** may communicate with the internal and external controller **311** to trigger seal actuators **311** to close seals and/or to signal vent actuators **343** to close vents of the vehicle. The ECU **301** may communicate with the battery isolation controller **313** to trigger the battery isolation unit to close or to isolate the battery or batteries of the vehicle. While sensor data indicating extreme substance accumulation are received, the ECU **301** may communicate with ignition control controller **315** to disable vehicle ignition, and then to enable ignition when the substance accumulation condition is over.

[0085] In addition to, or instead of, transmitting signals for taking such steps, at **716** the ECU **301** may communicate via network communications module **337** over network **363** with a mobile device **373** of the driver to report the status of the substance accumulation, including the height level and the composition of the substance, the forecasted rate of substance accumulation or change, weather forecast or weather report data, the state of vehicle protective mechanisms, including seals, vents, battery isolation unit and the adaptive suspension systems, and/or to request instructions regarding any of these protective actions. Communication to the mobile device **373** may also warn against entering the vehicle and/or opening doors/windows of the vehicle. Similarly, the electronic control unit ECU **301** may communicate with a server **371** of an emergency response organization, such as a fire department or private vehicle protection organization, and/or with tow truck operators, roadside assistance personnel, or automated roadside assistance notification centers, to provide such status reports and to request further instructions regarding protective actions to take for the vehicle. The network communications module **337** may be connected to a cellular mobile telephone carrier system via radio cell towers. In response, the server **371** and/or the mobile device **373** may transmit to the ECU **301** to implement, or to continue to implement some or all vehicle protection actions as programmed, or to stand down and to disable vehicle protection actions. Also, the server **371** and/or the mobile device **373** may transmit to the a

command ECU 301 to reverse some or all of the protective actions taken, for example, if a threat to the vehicle is deemed not be real, or when the threat is deemed to have passed.

[0086] The term “and/or,” may be understood to mean “either or both” of the elements thus indicated. Additional elements may optionally be present unless excluded by the context. Terms such as “first,” “second,” “third” in the claims referring to a structure, module or step should not necessarily be construed to mean precedence or temporal order but are generally intended to distinguish between claim elements.

[0087] The above-described embodiments are intended to be examples only. Components or processes described as separate may be combined or combined in ways other than as described, and components or processes described as being together or as integrated may be provided separately. Steps or processes described as being performed in a particular order may be re-ordered or recombined.

[0088] Features and limitations described in any one embodiment may be applied to any other embodiment herein, and flowcharts or examples relating to one embodiment may be combined with any other embodiment in a suitable manner, done in different orders, or done in parallel. In addition, the systems and methods described herein may be performed in real time.

[0089] It should also be noted that the systems and/or methods described above may be applied to, or used in accordance with, other systems and/or methods. In various embodiments, additional elements may be included, some elements may be removed, and/or elements may be arranged differently from what is shown. Alterations, modifications and variations can be affected to the particular embodiments by those of skill in the art without departing from the scope of the present application, which is defined solely by the claims appended hereto.

Claims

1. A vehicle wheel comprising: a first sensor positioned at a rim of the wheel at a first distance from a center of the wheel, wherein the first sensor comprises first probes configured to detect electrical conductivity of a substance in contact with the wheel; a second sensor positioned at the rim of the wheel at the first distance from the center of the wheel and spaced remote from the first sensor, wherein the second sensor comprises second probes configured to detect the electrical conductivity of the substance in contact with the wheel; and a wireless transmitter configured to transmit signals indicating the electrical conductivity detected by the first sensor and the electrical conductivity detected by the second sensor.
2. The vehicle wheel of claim 1, wherein the first sensor and the second sensor each comprise a temperature sensor.
3. The vehicle wheel of claim 1, further comprising an orientation sensor configured to detect an orientation of the wheel.
4. The vehicle wheel of claim 1, further comprising a third sensor positioned at the rim of the wheel at the first distance from the center of the wheel and spaced equidistant from the first sensor and from the second sensor, wherein the first sensor is positioned equidistant from the second sensor and from the third sensor, wherein the third sensor comprises third probes configured to detect the electrical conductivity of the substance in contact with the wheel.
5. The vehicle wheel of claim 4, wherein one probe of the first probes of the first sensor is positioned at an inner ring of the rim of the wheel, and another probe of the second probes of the second sensor is positioned at an outer ring of the rim of the wheel.
6. A system comprising the vehicle wheel of claim 1, further comprising: an onboard data processor positioned and configured to receive the signals indicating the electrical conductivity detected by the first sensor and the electrical conductivity detected by the second sensor, wherein the onboard data processor is configured to determine a protective action for the vehicle based on the signals received.

7. A method comprising: detecting, by a first sensor positioned at a rim of a wheel of a vehicle, electrical conductivity of a substance in contact with the wheel, wherein the first sensor is at a first distance from a center of the wheel and comprises first probes; detecting, by a second sensor positioned at the rim of the wheel, the electrical conductivity of the substance in contact with the wheel, wherein the second sensor is at the first distance from the center of the wheel and spaced remote from the first sensor, wherein the second sensor comprises second probes; transmitting, by a wireless transmitter, signals indicating the electrical conductivity detected by the first sensor and the electrical conductivity detected by the second sensor; receiving by an onboard data processor the signals indicating the electrical conductivity detected by the first sensor and the electrical conductivity detected by the second sensor; determining, by the onboard data processor, a protective action for the vehicle based on the signals received; and causing the onboard data processor to transmit signals to cause at least one system of the vehicle to take the protective action for the vehicle.

8. The method of claim 7, further comprising taking, by the at least one system of the vehicle, the protective action for the vehicle.

9. The method of claim 8, wherein the at least one system comprises an adjustable suspension.

10. The method of claim 7, wherein the protective action comprises lowering the vehicle when the substance in contact with the wheel comprises a solid.

11. The method of claim 10, wherein the substance in contact with the wheel comprises snow.

12. The method of claim 7, wherein the protective action comprises raising the vehicle when the substance in contact with the wheel comprises a liquid.

13. The method of claim 12, wherein the substance in contact with the wheel comprises water.

14. The method of claim 7, wherein the determining of the protective action for the vehicle is based at least in part on determining that the substance is in contact with the first sensor and with the second sensor.

15. The method of claim 7, wherein the determining of the protective action for the vehicle is based at least in part on determining a type of substance that is in contact with the wheel.

16. The method of claim 7, wherein the determining of the protective action for the vehicle is based at least in part on determining that the substance in contact with the wheel comprises liquid.

17. The method of claim 7, wherein the determining of the protective action for the vehicle is based at least in part on determining a protective ride height of the vehicle.

18. The method of claim 7, wherein the determining of the protective action for the vehicle is based at least in part on: determining, at a first time, that the substance is in contact with the first sensor and that the second sensor is free of contact with the substance; determining, at a second time subsequent to the first time, that the first sensor and the second sensor are both in contact with the substance.

19. The method of claim 18, further comprising determining an accumulation rate of the substance in contact with the wheel based on a time interval between the first time and the second time.

20. The method of claim 7, further comprising transmitting, to a device remote from the vehicle, a notification indicating substance accumulation at the vehicle.
